Program and Branch Name:
[BTech CSE-F]

PBL Progress Report –II

Field	Details
Course Name	Data Structure and Algorithms - II
Course Code	CCSE0301
Faculty Name	Mr. Shamshad Ali
Assignment Type	Individual Assignment
Student Name	Mohd Sabir Ansari
ERP ID	0261DCS032
Team/Group No. (Group/Capstone only)	Not Applicable - Individual Assignment
Problem Statement / Project Title	Intelligent Education Resource Sharing System
SDG(s)	SDG 4 — Quality Education , SDG 17 — Partnerships
Reporting Month	Month 2
Date of Submission	08/10/2026

Student Declaration

I certify that this assignment is my/our own work, with all sources appropriately acknowledged. I understand that a 0% mark and further academic penalties may apply if the content is found to be plagiarised or inappropriately AI-generated.

Student Name & Signature: Mohd Sabir Ansari &
Date: 08/10/2026

PBL Progress Report – II

Review 2: Development & Application

Review Coverage:

- **Individual / Group Assignment:** Unit/Module 2 and Unit/Module 3

<table>
<tr><th>Field</th><th>Details</th></tr>
<tr><td>1. Previous Review Feedback</td><td>

Focus more on DSA-II Unit 1 and Unit 2 concepts.
Analyze how Graph concepts can be used in the proposed system.
Start the implementation phase of the system.
Plan the implementation of the topics selected in the previous review.
Prepare the report with proper formatting and documentation.

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<tr><td>2. Action Taken on Previous Feedback</td><td>

Studied DSA-II Unit 1 and Unit 2 concepts in greater depth.
Analyzed the practical use of Graph concepts in the proposed system.
Started the implementation phase of the system.
Planned the implementation of the DSA topics selected in the previous review.
Improved the formatting and documentation of the project report.

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<tr><td>3. Research / Data Collection Progress</td><td>
<table>
<tr><th>Research Paper</th><th>Limitation / Gap</th><th>Our Project Contribution</th></tr>
<tr><td>DFRP: A Dual-Track Feedback Recommendation System for Educational Resources</td><td>Complex knowledge graph and recommendation-based approach.</td><td>Simpler DSA-based resource searching, exploration and organization.</td></tr>
<tr><td>Learning Resource Sharing Platform Based on Mobile Interactive Technology</td><td>Complex deep-learning and knowledge-graph architecture.</td><td>Lightweight algorithm-based resource-sharing approach.</td></tr>
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<tr><td>4. Unit/Module Concepts Applied</td><td>
<table>
<tr><th>Concept</th><th>Description</th><th>Time Complexity</th></tr>
<tr><td>Graph</td><td>Represents relationships among students, subjects, topics, and educational resources for efficient resource discovery and exploration.</td><td>O(V + E)</td></tr>
<tr><td>BFS (Breadth First Search)</td><td>Level-by-level discovery of educational resources from a subject/topic.</td><td>O(V + E)</td></tr>
<tr><td>DFS (Depth First Search)</td><td>Deep exploration of resources connected to a selected subject/topic.</td><td>O(V + E)</td></tr>
<tr><td>Kruskal’s Algorithm</td><td>Creates an efficient minimum-cost connection network among related educational resources.</td><td>O(E log E)</td></tr>
<tr><td>Dijkstra’s Algorithm</td><td>Finds the least-cost/best path from a student/resource to a target educational resource.</td><td>O((V + E) log V)</td></tr>
<tr><td>0/1 Knapsack</td><td>Selects the most useful resources within limited study time or storage capacity.</td><td>O(nW)</td></tr>
</table>
<table>
<tr><th>Concept Not Used</th><th>Reason</th></tr>
<tr><td>Prim’s Algorithm</td><td>Kruskal’s Algorithm provides the required minimum-cost network functionality; using both would be redundant.</td></tr>
<tr><td>Bellman-Ford Algorithm</td><td>Negative edge weights are not required for resource relationships, so Dijkstra’s Algorithm is more suitable.</td></tr>
<tr><td>Floyd-Warshall Algorithm</td><td>All-pairs shortest paths are unnecessary for the current system scope.</td></tr>
<tr><td>Longest Common Subsequence (LCS)</td><td>Comparing common sequences is not a core requirement of the resource-sharing system.</td></tr>
<tr><td>Matrix Chain Multiplication</td><td>Matrix multiplication optimization has no direct requirement in the current system.</td></tr>
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	Resource Allocation Problem	Resource selection can be handled through 0/1 Knapsack; a separate allocation model is not required.
5. Methodology / Approach Followed	We followed a **Quantitative Research Methodology** with a practical DSA-based analytical approach. First, the educational resources were modelled using Trees and Graphs to define their organization and relationships. Then, the selected DSA concepts were analysed and mapped to specific system requirements such as resource searching, exploration, path finding, and resource selection. The approach focuses on applying suitable algorithms to real system problems and evaluating their efficiency using measurable factors such as time complexity and performance.	
6. Work Completed During Month 2	• Studied DSA-II Unit 1 and Unit 2 concepts in greater depth. • Started the implementation phase of the concepts selected from Unit 1 and Unit 2. • Studied Unit 3 and Unit 4 concepts and identified the algorithms relevant to the proposed system. • Selected useful algorithms such as BFS, DFS, Kruskal's, Dijkstra's, and 0/1 Knapsack based on their system applications. • Planned the implementation of the selected algorithms for resource searching, exploration, path finding, and resource selection.	
7. Analysis / Development / Implementation	• Developed the Tree-based structure for hierarchical organization of educational resources. • Designed the Graph-based model to represent relationships between students, subjects, topics, and resources. • Started implementation of the selected Unit 1 and Unit 2 concepts in the proposed system. • Analysed the selected Unit 3 and Unit 4 algorithms and planned their implementation according to system requirements.	
8. Results / Interim Outcomes	The relevance of each DSA concept to the proposed system is now clearly understood. We have identified which concepts will be used for specific system functionalities and which are not required for the current system scope.	
9. Evidence / Supporting Material	GitHub Repository - https://github.com/sabiransari786/Intelligent-Eductation-Resource-sharing-system	
10. Challenges & Corrective Actions	• **Concept-to-System Mapping:** Understanding where and how each DSA concept could be applied within the proposed system was a major challenge. • **Implementation Planning:** Determining how the selected concepts would be implemented and integrated into the appropriate system functionalities was challenging. • **Resource organization:** Designed a clear Tree hierarchy for department, course, semester, subject, unit, topic, and educational resources. • **Graph representation:** Defined meaningful relationships between students, subjects, topics, and resources.	
11. Individual / Team Contribution	Independent contribution included problem analysis, literature and research paper review, DSA concept analysis and selection, mapping of DSA concepts with system functionalities, system design, implementation planning, documentation, and testing preparation for the Intelligent Education Resource Sharing System. These activities supported the overall development and refinement of the proposed system.	
12. Overall Progress (%)	**60%**	
13. Plan for Final Review	• Complete the implementation of the DSA concepts selected in Review Report II. • Study the remaining units and select relevant topics applicable to the proposed system. • Plan and complete the implementation of the newly selected concepts. • Integrate all selected concepts and develop the final working system. • Complete the detailed project documentation and final report.	

Faculty Evaluation:

- To be completed by Faculty in Moodle